

Math 411 Summer 2025**Midterm 2****August 22, 2025****8:30 - 9:30 AM**

Name: _____

Student ID: _____

Instructions:

1. Be cool. You can do it!
2. You may only work on your own.
3. Please try to write as legible as you can.
4. Do not write on the back of any page.
5. Please try to provide as many detail as you can.
6. Write down and sign your honor statement.

Question:	1	2	3	Total
Points:	5	10	15	30
Score:				

1. Determine if the following is true/false and give an explanation/a counterexample.

(a) (2 points) Is $\mathbb{Z}_2[x] \bmod x^2 + 1$ a field?

(b) (1 point) Is $\mathbb{Z}_3[x] \bmod x^2 + 1$ is a field?

(c) (2 points) Is $\mathbb{Z}_5[x] \bmod x^2 + 1$ a field?

2. Consider $\mathbb{Z}_p[x] \bmod x^2 + 1$ for some odd prime number $p \in \mathbb{N}$.

(a) (2 points) Show that $p \equiv 1 \pmod{4}$ or $p \equiv 3 \pmod{4}$ and deduce $\frac{p-1}{2}$ is always an integer.

(b) (4 points) Suppose there exists $x \in \mathbb{Z}_p$ such that $x^2 \equiv -1 \pmod{p}$. Fermat's little theorem says $a^{p-1} \equiv 1 \pmod{p}$ for $0 \neq a \in \mathbb{Z}_p$. Note $(-1)^{\frac{p-1}{2}} \equiv (x^2)^{\frac{p-1}{2}}$. Show that $p \equiv 1 \pmod{4}$.

(c) (4 points) Deduce that if $p \equiv 3 \pmod{4}$, then $\mathbb{Z}_p[x] \bmod x^2 + 1$ is a field.

3. Let $K \subseteq L$ be two fields and $\gamma \in L$. We say γ is algebraic over K if there exists some non-zero polynomial $f(x) \in K[x]$ such that γ is a root of $f(x)$. For instance, $\sqrt{2} \in \mathbb{R}$ is algebraic over $\mathbb{Q}$ because it is a root of $x^2 - 2$ which is in $\mathbb{Q}[x]$. Show the following.
- (a) (2 points) Consider $K = \mathbb{Q}$ and $L = \mathbb{R}$. Is $\sqrt{2} + 1 \in L$ algebraic over K ?

From now on, let $\gamma \in L$ be algebraic over K and $A = \{f(x) \in K[x] : f(\gamma) = 0\}$.

- (b) (1 point) Show that A is a non-empty set.

(c) (6 points) Show there exists some irreducible polynomial $m(x) \in A$ such that $A = (m(x))$.

We may further assume the leading coefficient of $m(x)$ to be 1 since K is field. We denote this element by $m_\gamma(x)$ and call it the minimal polynomial of γ in $K[x]$.

- (d) (6 points) Let $K = \mathbb{Q}$ and $L = \mathbb{R}$. Let $\gamma = \sqrt{2 + \sqrt{2}}$. Find $m_\gamma(x) \in K[x]$ and explain why it is irreducible in $K[x]$.

Scratch paper. If you write any answer here, make sure you label the question clearly.

Scratch paper. If you write any answer here, make sure you label the question clearly.